

数学物理方法作业

1900011604 张植竣 北京大学

第七章

1.(1) 对于 $f(z) = \frac{e^{z^2}}{z-1} = \frac{P(z)}{Q(z)}$, $P(1) = e \neq 0$, $Q(1) = 0$, $Q'(1) = 1 \neq 0$, 则 $\operatorname{res}f(1) = \frac{P(1)}{Q'(1)} = e$.

1.(3) 对于 $f(z) = \frac{z^2}{(1-\cos z)^2} = \frac{P(z)}{Q(z)}$, $z_0 = 0$ 是 $P(z)$ 的二阶零点, 是 $Q(z)$ 的四阶零点, 则 $z_0 = 0$ 是 $f(z)$ 的二阶极点:

$$\operatorname{res}f(0) = \lim_{z \rightarrow 0} \left[\frac{d}{dz} z^2 \frac{z^2}{(1-\cos z)^2} \right] = \lim_{z \rightarrow 0} \frac{-2z^4 \sin z}{1-\cos z} = 0$$

最后一步使用了两次 l'Hospital 法则.

1.(5) 对于 $f(z) = \frac{e^z}{(z^2-1)^2} = \frac{P(z)}{Q(z)}$, $z_0 = 1$ 是 $P(z)$ 的解析点, 是 $Q(z)$ 的二阶零点, 则:

$$\operatorname{res}f(1) = \frac{1}{(2-1)!} \lim_{z \rightarrow 1} \left[\frac{d}{dz} (z-1)^2 \frac{e^z}{(z^2-1)^2} \right] = \lim_{z \rightarrow 1} \frac{e^z(z-1)}{(z+1)^3} = 0$$

2.(1) 对于 $f(z) = \frac{1}{z^3-z^5} = \frac{1}{z^3(1+z)(1-z)}$, $z = 0$ 是 $f(z)$ 的三阶零点, $z = 1$ 和 $z = -1$ 是 $f(z)$ 的一阶零点.

对于 $z = 0$,

$$\operatorname{res}f(0) = \frac{1}{(3-1)!} \lim_{z \rightarrow 0} \left[\frac{d^2}{dz^2} \left(z^3 \frac{1}{z^3-z^5} \right) \right] = \frac{1}{2} \lim_{z \rightarrow 0} \left(\frac{1}{(1-z)^3} + \frac{1}{(1+z)^3} \right) = 1$$

对于 $z = 1$, $f(z) = \frac{1}{z^3-z^5} = \frac{P(z)}{Q(z)}$, $P(1) = 1 \neq 0$, $Q(1) = 0$, $Q'(1) = -2 \neq 0$, 则 $\operatorname{res}f(1) = \frac{P(1)}{Q'(1)} = -\frac{1}{2}$.

对于 $z = -1$, $f(z) = \frac{1}{z^3-z^5} = \frac{P(z)}{Q(z)}$, $P(-1) = 1 \neq 0$, $Q(-1) = 0$, $Q'(-1) = -2 \neq 0$, 则 $\operatorname{res}f(-1) = \frac{P(-1)}{Q'(-1)} = -\frac{1}{2}$.

2.(3) 对于 $f(z) = \frac{z}{1-\cos z}$, 函数的奇点为 $z_0 = 2k\pi$ ($k \in \mathbb{Z}$).

$k = 0$ 时, z_0 为函数的一阶极点, 则:

$$\operatorname{res}f(z_0) = \lim_{z \rightarrow z_0} \left[z \left(\frac{z}{1-\cos z} \right) \right] = \lim_{z \rightarrow 0} \frac{z^2}{1-\cos z} = 2$$

最后一步使用了 l'Hospital 法则.

$k \neq 0$ 时, z_0 为函数的二阶极点, 则求函数在该点的 Laurent 展开.

令 $t = z - z_0$, 则 $\frac{z}{1-\cos z} = \frac{t+z_0}{1-\cos t} = \sum_{n=-2}^{\infty} c_n t^n$, 于是有:

$$t + z_0 = \left(\frac{1}{2}t^2 - \frac{1}{24}t^4 + \cdots \right) (c_{-2}t^{-2} + c_{-1}t^{-1} + c_0t^0 + \cdots)$$

比较一次项系数: $\frac{1}{2}t^2 \times c_{-1}t^{-1} = t$, 即得 $\operatorname{res}f(z_0) = c_{-1} = 2$.

综上所述, 函数在奇点的留数为:

$$\operatorname{res}f(2k\pi) = 2 \quad (k \in \mathbb{Z})$$

2.(5) 对于 $f(z) = e^{\frac{1}{2}(z-\frac{1}{z})}$, 函数的奇点为 $z_1 = 0, z_2 = \infty$.

求 $f(z)$ 在 $z=0$ 处的 Laurent 展开 $\sum_{n=-\infty}^{\infty} c_n t^n$ 如下:

$$e^{\frac{1}{2}(z-\frac{1}{z})} = \sum_{n=-2}^{\infty} \frac{1}{n!} \frac{1}{2^n} \left(z - \frac{1}{z}\right)^n = \frac{1}{0!} \frac{1}{2^0} \left(z - \frac{1}{z}\right)^0 + \frac{1}{1!} \frac{1}{2^1} \left(z - \frac{1}{z}\right)^1 + \frac{1}{2!} \frac{1}{2^2} \left(z - \frac{1}{z}\right)^2 + \cdots$$

$$\operatorname{res} f(0) = c_{-1} = \sum_{n=0}^{\infty} \frac{1}{(2n+1)!} \frac{1}{2^{2n+1}} (-1)^{n+1} \binom{n}{2n+1} = -J_1(1)$$

最后一步用到了 $\binom{n}{2n+1} = \frac{(2n+1)!}{n!(n+1)!}$ 和 $\Gamma(n+2) = (n+1)!$.

则 $\operatorname{res} f(\infty) = -\operatorname{res} f(0) = J_1(1)$

2.(7) 对于 $f(z) = \frac{1}{(z-1)\ln z}$, 因内含多值函数 $\ln z$, 分类讨论如下:

若取单值分支 $\arg 1 = 0$, 则函数有二阶极点 $z = 1, f(z)$ 在该点的留数为:

$$\operatorname{res} f(1) = \lim_{z \rightarrow 1} \left[\frac{d}{dz} (z-1)^2 \frac{1}{(z-1)\ln z} \right] = \lim_{z \rightarrow 1} \frac{z \ln z - (z-1)}{z \ln^2 z} = \frac{1}{2}$$

最后一步使用了两次 l'Hospital 法则.

若取单值分支 $\arg 1 = 2k\pi$ ($k \in \mathbb{Z} \setminus \{0\}$), 则函数有一阶极点 $z = 1, f(z)$ 在该点的留数为:

$$\operatorname{res} f(1) = \lim_{z \rightarrow 1} (z-1)f(z) = \frac{1}{2k\pi i} \quad (k \in \mathbb{Z} \setminus \{0\})$$

4.(1) 在积分围道 $|z-1|=1$ 内, 函数 $\frac{1}{1+z^4}$ 有两个奇点 $z_1 = e^{\frac{i\pi}{4}}$ 和 $z_2 = e^{\frac{-i\pi}{4}}$.

对于 $f(z) = \frac{1}{1+z^4} = \frac{P(z)}{Q(z)}$, $P(z_1) = 1 \neq 0$, $Q(z_1) = 0$, $Q'(z_1) = -2\sqrt{2} + 2\sqrt{2}i \neq 0$, 则 $\operatorname{res} f(z_1) = \frac{P(z_1)}{Q'(z_1)} = -\frac{1+i}{4\sqrt{2}}$. 同理可得 $\operatorname{res} f(z_2) = -\frac{1-i}{4\sqrt{2}}$. 故:

$$\oint_{|z-1|=1} \frac{1}{1+z^4} = 2\pi i \left(-\frac{1+i}{4\sqrt{2}} - \frac{1-i}{4\sqrt{2}} \right) = -\frac{\pi i}{\sqrt{2}}$$

4.(3) 在积分围道 $|z-1|=1$ 内, 函数 $\frac{1}{z^2-1} \sin \frac{\pi z}{4}$ 有一个奇点 $z = 1$.

对于 $f(z) = \frac{1}{z^2-1} \sin \frac{\pi z}{4} = \frac{P(z)}{Q(z)}$, $P(1) = \frac{\sqrt{2}}{2} \neq 0$, $Q(1) = 0$, $Q'(1) = 2 \neq 0$, 则 $\operatorname{res} f(1) = \frac{P(1)}{Q'(1)} = \frac{\sqrt{2}}{4}$. 故:

$$\oint_{|z-1|=1} \frac{1}{z^2-1} \sin \frac{\pi z}{4} = 2\pi i \frac{\sqrt{2}}{4} = \frac{\sqrt{2}}{2} \pi i$$

4.(5) 在积分围道 $|z|=n$ 内, 函数 $\tan \pi z$ 有 $2n$ 个奇点: $z_k = \frac{1}{2} + k$ ($k \in \mathbb{Z}$, $-n+1 \leq k \leq n-1$).

对于 $f(z) = \tan \pi z = \frac{\sin \pi z}{\cos \pi z} = \frac{P(z)}{Q(z)}$, $P(z_k) = \pm 1 \neq 0$, $Q(z_k) = 0$, $Q'(z_k) = \mp \pi \neq 0$, 则 $\operatorname{res} f(z_k) = \frac{P(z_k)}{Q'(z_k)} = -\frac{1}{\pi}$. 故:

$$\oint_{|z|=n} \tan \pi z = 2\pi i \sum_{k=-n+1}^{n-1} \left(-\frac{1}{\pi} \right) = -4ni$$

4.(7) 在积分围道 $|z| = 1$ 内, 函数 $\frac{e^z}{z^3}$ 有一个三阶极点 $z = 0$. 函数在该处的留数为:

$$\operatorname{res} f(0) = \frac{1}{(3-1)!} \lim_{z \rightarrow 0} \left[\frac{d^2}{dz^2} \left(z^3 \frac{e^z}{z^3} \right) \right] = \frac{1}{2} \lim_{z \rightarrow 0} e^z = \frac{1}{2}$$

故:

$$\oint_{|z|=1} \frac{e^z}{z^3} = 2\pi i \times \frac{1}{2} = \pi i$$

5.(1) 作变换 $z = e^{i\theta}$, 则:

$$\int_0^{2\pi} \cos^{2n} \theta d\theta = \oint_{|z|=1} \left(\frac{z^2 + 1}{2z} \right)^{2n} \frac{dz}{iz} = 2\pi \sum_{|z|<1} \operatorname{res} f(z)$$

其中:

$$f(z) = \frac{1}{z} \left(\frac{z^2 + 1}{2z} \right)^{2n} = \frac{(z^2 + 1)^{2n}}{2^{2n} z^{2n+1}}$$

在积分围道 $|z| = 1$ 内, $f(z)$ 有一个 $2n + 1$ 阶极点 $z = 0$, 函数在该点处的留数为:

$$\begin{aligned} \operatorname{res} f(0) &= \frac{1}{(2n)!} \lim_{z \rightarrow 0} \left[\frac{d^{2n}}{dz^{2n}} z^{2n+1} f(z) \right] \\ &= \frac{1}{(2n)! 2^{2n}} \lim_{z \rightarrow 0} \left[\frac{d^{2n}}{dz^{2n}} (z^2 + 1)^{2n} \right] \\ &= \frac{1}{(2n)! 2^{2n}} \lim_{z \rightarrow 0} \left[\frac{d^{2n}}{dz^{2n}} \sum_{k=0}^{2n} \binom{2n}{k} z^{2k} \right] \\ &= \frac{1}{(2n)! 2^{2n}} \lim_{z \rightarrow 0} \left[\frac{d^{2n}}{dz^{2n}} \binom{2n}{n} z^{2n} \right] \\ &= \frac{1}{(2n)! 2^{2n}} \binom{2n}{n} (2n)! \\ &= \frac{(2n)!}{2^{2n} (n!)^2} \end{aligned}$$

故:

$$\begin{aligned} \int_0^{2\pi} \cos^{2n} \theta d\theta &= \oint_{|z|=1} \left(\frac{z^2 + 1}{2z} \right)^{2n} \frac{dz}{iz} \\ &= 2\pi \sum_{|z|<1} \operatorname{res} f(z) \\ &= 2\pi \frac{(2n)!}{2^{2n} (n!)^2} \\ &= 2\pi \frac{(2n-1)!!}{(2n)!!} \end{aligned}$$

5.(3) 作变换 $z = e^{i\theta}$, 由 $\sin(\theta + \pi) = \sin \theta$ 得:

$$\begin{aligned} \int_0^\pi \frac{d\theta}{1 + \sin^2 \theta} &= \frac{1}{2} \int_0^{2\pi} \frac{d\theta}{1 + \sin^2 \theta} \\ &= \frac{1}{2} \oint_{|z|=1} \frac{4iz}{z^4 - 6z + 1} dz \\ &= 2i \oint_{|z|=1} \frac{z}{z^4 - 6z + 1} dz \end{aligned}$$

对于 $f(z) = \frac{z}{z^4 - 6z + 1} = \frac{P(z)}{Q(z)}$, 函数有四个一阶极点: $z_1 = \sqrt{3 - 2\sqrt{2}}$, $z_2 = -\sqrt{3 - 2\sqrt{2}}$, $z_3 = \sqrt{3 + 2\sqrt{2}}$, $z_4 = -\sqrt{3 + 2\sqrt{2}}$.

在积分围道 $|z| = 1$ 内, $f(z)$ 有两个一阶极点 z_1, z_2 .

对于 $z_1 = \sqrt{3 - 2\sqrt{2}}$, $P(z_1) = \sqrt{3 - 2\sqrt{2}} \neq 0$, $Q(z_1) = 0$, $Q'(z_1) = -8\sqrt{2} \neq 0$, 则 $\operatorname{res}f(z_1) = \frac{P(z_1)}{Q'(z_1)} = -\frac{1}{8\sqrt{2}}$. 同理可得 $\operatorname{res}f(z_2) = -\frac{1}{8\sqrt{2}}$.

故:

$$\begin{aligned} \int_0^\pi \frac{d\theta}{1 + \sin^2 \theta} &= \frac{1}{2} \int_0^{2\pi} \frac{d\theta}{1 + \sin^2 \theta} \\ &= \frac{1}{2} \oint_{|z|=1} \frac{4iz}{z^4 - 6z + 1} dz \\ &= 2i \oint_{|z|=1} \frac{z}{z^4 - 6z + 1} dz \\ &= 2i \times 2\pi i \sum_{|z|<1} \operatorname{res}f(z) \\ &= 2i \times 2\pi i \left(-\frac{1}{8\sqrt{2}} - \frac{1}{8\sqrt{2}} \right) \\ &= \frac{\pi}{\sqrt{2}} \end{aligned}$$

6.(1) 考虑积分: $\oint_C \frac{z^2}{1+z^4} dz$, 积分围道 C 如下:

从 $z = -R$ 沿实轴到 $z = R$, 再沿 $|z| = R$, $0 < \arg z < \pi$ 的半圆 C_R 逆时针回到 $z = -R$. 即:

$$\oint_C \frac{z^2}{1+z^4} dz = \int_{-R}^R \frac{z^2}{1+z^4} dz + \int_{C_R} \frac{z^2}{1+z^4} dz$$

取 $R \rightarrow \infty$ 的极限, 则有:

$$\int_{-\infty}^{\infty} \frac{x^2}{1+x^4} dx = \int_{-\infty}^{\infty} \frac{z^2}{1+z^4} dz = \lim_{R \rightarrow \infty} \oint_C \frac{z^2}{1+z^4} dz - \lim_{R \rightarrow \infty} \int_{C_R} \frac{z^2}{1+z^4} dz$$

由大圆弧定理:

$$\lim_{R \rightarrow \infty} \int_{C_R} \frac{z^2}{1+z^4} dz = 0$$

故:

$$\int_{-\infty}^{\infty} \frac{x^2}{1+x^4} dx = \lim_{R \rightarrow \infty} \oint_C \frac{z^2}{1+z^4} dz = 2\pi i \sum_{\operatorname{Im} z > 0} \operatorname{res}f(z)$$

函数 $f(z) = \frac{z^2}{1+z^4}$, 在 $\operatorname{Im} z > 0$ 的区域内有两个奇点 $z_1 = e^{\frac{i\pi}{4}}$ 和 $z_2 = e^{\frac{i3\pi}{4}}$.

对于 $f(z) = \frac{z^2}{1+z^4} = \frac{P(z)}{Q(z)}$, $P(z_1) = e^{\frac{i\pi}{2}} \neq 0$, $Q(z_1) = 0$, $Q'(z_1) = 4e^{\frac{i3\pi}{4}} \neq 0$, 则 $\operatorname{res}f(z_1) = \frac{P(z_1)}{Q'(z_1)} = \frac{1}{4}e^{\frac{-i\pi}{4}}$. 同理可得 $\operatorname{res}f(z_2) = \frac{1}{4}e^{\frac{-i3\pi}{4}}$. 故:

$$\int_{-\infty}^{\infty} \frac{x^2}{1+x^4} dx = \lim_{R \rightarrow \infty} \oint_C \frac{z^2}{1+z^4} dz = 2\pi i \left(\frac{1}{4}e^{\frac{-i\pi}{4}} + \frac{1}{4}e^{\frac{-i3\pi}{4}} \right) = \frac{\pi}{\sqrt{2}}$$

6.(3) 考虑积分: $\oint_C \frac{1}{(1+z^2)^{n+1}} dz$, 积分围道 C 如下:

从 $z = -R$ 沿实轴到 $z = R$, 再沿 $|z| = R$, $0 < \arg z < \pi$ 的半圆 C_R 逆时针回到 $z = -R$. 即:

$$\oint_C \frac{1}{(1+z^2)^{n+1}} dz = \int_{-R}^R \frac{1}{(1+z^2)^{n+1}} dz + \int_{C_R} \frac{1}{(1+z^2)^{n+1}} dz$$

取 $R \rightarrow \infty$ 的极限, 则有:

$$\int_{-\infty}^{\infty} \frac{1}{(1+x^2)^{n+1}} dx = \int_{-\infty}^{\infty} \frac{1}{(1+z^2)^{n+1}} dz = \lim_{R \rightarrow \infty} \oint_C \frac{1}{(1+z^2)^{n+1}} dz - \lim_{R \rightarrow \infty} \int_{C_R} \frac{1}{(1+z^2)^{n+1}} dz$$

由大圆弧定理:

$$\lim_{R \rightarrow \infty} \int_{C_R} \frac{1}{(1+z^2)^{n+1}} dz = 0$$

故:

$$\int_{-\infty}^{\infty} \frac{1}{(1+z^2)^{n+1}} dx = \lim_{R \rightarrow \infty} \oint_C \frac{1}{(1+z^2)^{n+1}} dz = 2\pi i \sum_{\operatorname{Im} z > 0} \operatorname{res} f(z)$$

在 $\operatorname{Im} z > 0$ 的区域内, $f(z) = \frac{1}{(1+z^2)^{n+1}}$ 有一个 $n+1$ 阶极点 $z = i$, 函数在该点处的留数为:

$$\begin{aligned} \operatorname{res} f(i) &= \frac{1}{n!} \lim_{z \rightarrow i} \left[\frac{d^n}{dz^n} (z-i)^{n+1} f(z) \right] \\ &= \frac{1}{n!} \lim_{z \rightarrow i} \left[\frac{d^{2n}}{dz^{2n}} \frac{1}{(z+i)^{n+1}} \right] \\ &= \frac{1}{n!} \lim_{z \rightarrow i} \left[(-1)^n \frac{(2n)!}{n!} (z+i)^{-2n-1} \right] \\ &= \frac{1}{n!} (-1)^n \frac{(2n)!}{n!(2i)^{2n+1}} \\ &= \frac{(2n)!}{2^{2n}(n!)^2} \times \frac{1}{2i} \\ &= \frac{(2n-1)!!}{(2n)!!} \times \frac{1}{2i} \end{aligned}$$

故:

$$\int_{-\infty}^{\infty} \frac{1}{(1+z^2)^{n+1}} dx = \lim_{R \rightarrow \infty} \oint_C \frac{1}{(1+z^2)^{n+1}} dz = 2\pi i \times \frac{(2n-1)!!}{(2n)!!} \times \frac{1}{2i} = \pi \frac{(2n-1)!!}{(2n)!!}$$

7.(1) 考虑积分: $\oint_C \frac{e^{iz}}{1+z^4} dz$, 积分围道 C 如下:

从 $z = -R$ 沿实轴到 $z = R$, 再沿 $|z| = R$, $0 < \arg z < \pi$ 的半圆 C_R 逆时针回到 $z = -R$. 即:

$$\oint_C \frac{e^{iz}}{1+z^4} dz = \int_{-R}^R \frac{e^{iz}}{1+z^4} dz + \int_{C_R} \frac{e^{iz}}{1+z^4} dz$$

取 $R \rightarrow \infty$ 的极限, 则有:

$$\int_{-\infty}^{\infty} \frac{e^{iz}}{1+z^4} dz = \lim_{R \rightarrow \infty} \oint_C \frac{e^{iz}}{1+z^4} dz - \lim_{R \rightarrow \infty} \int_{C_R} \frac{e^{iz}}{1+z^4} dz$$

由 Jordan 引理:

$$\lim_{R \rightarrow \infty} \int_{C_R} \frac{e^{iz}}{1+z^4} dz = 0$$

故:

$$\int_{-\infty}^{\infty} \frac{e^{iz}}{1+z^4} dz = \lim_{R \rightarrow \infty} \oint_C \frac{e^{iz}}{1+z^4} dz = 2\pi i \sum_{\text{Im } z > 0} \text{res} f(z)$$

函数 $f(z) = \frac{e^{iz}}{1+z^4}$, 在 $\text{Im } z > 0$ 的区域内有两个奇点 $z_1 = e^{\frac{i\pi}{4}} = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$ 和 $z_2 = e^{\frac{i3\pi}{4}} = -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$.

对于 $f(z) = \frac{e^{iz}}{1+z^4} = \frac{P(z)}{Q(z)}$, $P(z_1) = e^{-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i} \neq 0$, $Q(z_1) = 0$, $Q'(z_1) = -2\sqrt{2} + 2\sqrt{2}i \neq 0$, 则:

$$\begin{aligned} \text{res} f(z_1) &= \frac{P(z_1)}{Q'(z_1)} \\ &= \frac{e^{-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i}}{-2\sqrt{2} + 2\sqrt{2}i} \\ &= e^{-\frac{\sqrt{2}}{2}} \left(\cos \frac{\sqrt{2}}{2} + i \sin \frac{\sqrt{2}}{2} \right) \left(-\frac{1+i}{2} \times \frac{1}{2\sqrt{2}} \right) \\ &= -\frac{\sqrt{2}}{8} e^{-\frac{\sqrt{2}}{2}} (1+i) \left(\cos \frac{\sqrt{2}}{2} + i \sin \frac{\sqrt{2}}{2} \right) \\ &= -\frac{\sqrt{2}}{8} e^{-\frac{\sqrt{2}}{2}} \left(\cos \frac{\sqrt{2}}{2} + i \sin \frac{\sqrt{2}}{2} + i \cos \frac{\sqrt{2}}{2} - \sin \frac{\sqrt{2}}{2} \right) \\ &= \frac{\sqrt{2}}{8} e^{-\frac{\sqrt{2}}{2}} \left(\sin \frac{\sqrt{2}}{2} - \cos \frac{\sqrt{2}}{2} - i \sin \frac{\sqrt{2}}{2} - i \cos \frac{\sqrt{2}}{2} \right) \end{aligned}$$

同理可得

$$\text{res} f(z_2) = \frac{\sqrt{2}}{8} e^{-\frac{\sqrt{2}}{2}} \left(\cos \frac{\sqrt{2}}{2} - i \sin \frac{\sqrt{2}}{2} - i \cos \frac{\sqrt{2}}{2} - \sin \frac{\sqrt{2}}{2} \right)$$

由 $\frac{\cos x}{1+x^4}$ 为偶函数得:

$$\begin{aligned} \int_0^{\infty} \frac{\cos x}{1+x^4} dx &= \frac{1}{2} \int_{-\infty}^{\infty} \frac{\cos x}{1+x^4} dx \\ &= \text{Re} \int_{-\infty}^{\infty} \frac{e^{iz}}{1+z^4} dz \\ &= \text{Re} \lim_{R \rightarrow \infty} \oint_C \frac{e^{iz}}{1+z^4} dz \\ &= \text{Re} \left\{ 2\pi i \left[\frac{\sqrt{2}}{8} e^{-\frac{\sqrt{2}}{2}} (-2i) \left(\cos \frac{\sqrt{2}}{2} + \sin \frac{\sqrt{2}}{2} \right) \right] \right\} \\ &= \frac{\sqrt{2}}{4} \pi e^{-\frac{\sqrt{2}}{2}} \left(\cos \frac{\sqrt{2}}{2} + \sin \frac{\sqrt{2}}{2} \right) \\ &= \frac{\pi}{2} e^{-\frac{\sqrt{2}}{2}} \sin \left(\frac{\sqrt{2}}{2} + \frac{\pi}{4} \right) \end{aligned}$$

7.(3) 考虑积分: $\oint_C \frac{ze^{iz}}{z^2-2z+2} dz$, 积分围道 C 如下:

从 $z = -R$ 沿实轴到 $z = R$, 再沿 $|z| = R$, $0 < \arg z < \pi$ 的半圆 C_R 逆时针回到 $z = -R$. 即:

$$\oint_C \frac{ze^{iz}}{z^2-2z+2} dz = \int_{-R}^R \frac{ze^{iz}}{z^2-2z+2} dz + \int_{C_R} \frac{ze^{iz}}{z^2-2z+2} dz$$

取 $R \rightarrow \infty$ 的极限, 则有:

$$\int_{-\infty}^{\infty} \frac{ze^{iz}}{z^2 - 2z + 2} dz = \lim_{R \rightarrow \infty} \oint_C \frac{ze^{iz}}{z^2 - 2z + 2} dz - \lim_{R \rightarrow \infty} \int_{C_R} \frac{ze^{iz}}{z^2 - 2z + 2} dz$$

由 Jordan 引理:

$$\lim_{R \rightarrow \infty} \int_{C_R} \frac{ze^{iz}}{z^2 - 2z + 2} dz = 0$$

故:

$$\int_{-\infty}^{\infty} \frac{ze^{iz}}{z^2 - 2z + 2} dz = \lim_{R \rightarrow \infty} \oint_C \frac{ze^{iz}}{z^2 - 2z + 2} dz = 2\pi i \sum_{\text{Im } z > 0} \text{res} f(z)$$

函数 $f(z) = \frac{ze^{iz}}{z^2 - 2z + 2}$, 在 $\text{Im } z > 0$ 的区域内有一个奇点 $1 + i$.

对于 $f(z) = \frac{ze^{iz}}{z^2 - 2z + 2} = \frac{P(z)}{Q(z)}$, $P(1 + i) = (1 + i)e^{-1+i} \neq 0$, $Q(1 + i) = 0$, $Q'(1 + i) = 2i \neq 0$, 则:

$$\begin{aligned} \text{res} f(1 + i) &= \frac{P(1 + i)}{Q'(1 + i)} \\ &= \frac{(1 + i)e^{-1+i}}{2i} \\ &= \frac{1}{2ei}(1 + i)(\cos 1 + i \sin 1) \\ &= \frac{1}{2ei}(\cos 1 + i \sin 1 + i \cos 1 - \sin 1) \end{aligned}$$

故:

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{x \sin x}{x^2 - 2x + 2} dx &= \text{Im} \int_{-\infty}^{\infty} \frac{ze^{iz}}{z^2 - 2z + 2} dz \\ &= \text{Im} \lim_{R \rightarrow \infty} \oint_C \frac{ze^{iz}}{z^2 - 2z + 2} dz \\ &= \text{Im} \left\{ 2\pi i \left[\frac{1}{2ei}(\cos 1 + i \sin 1 + i \cos 1 - \sin 1) \right] \right\} \\ &= \text{Im} \left\{ \frac{\pi}{e}(\cos 1 + i \sin 1 + i \cos 1 - \sin 1) \right\} \\ &= \frac{\pi}{e}(\sin 1 + \cos 1) \end{aligned}$$

8.(1) 记 $f(z) = \frac{1}{z(z-1)(z-2)}$.

考虑积分: $\oint_C f(z) dz$, 积分围道 C 如下:

从 $z = -R$ 沿实轴向右, 经过三段半圆 C_{δ_1} , C_{δ_2} , C_{δ_3} 绕过三个一阶极点 $z_1 = 0$, $z_2 = 1$, $z_3 = 2$ 到 $z = R$, 再沿 $|z| = R$, $0 < \arg z < \pi$ 的半圆 C_R 逆时针回到 $z = -R$. 即:

$$\begin{aligned} \oint_C f(z) dz &= \int_{-R}^{-\delta_1} f(z) dz + \int_{\delta_1}^{1-\delta_2} f(z) dz + \int_{1+\delta_2}^{2-\delta_3} f(z) dz + \int_{2+\delta_3}^R f(z) dz \\ &\quad + \int_{C_R} f(z) dz + \int_{C_{\delta_1}} f(z) dz + \int_{C_{\delta_2}} f(z) dz + \int_{C_{\delta_3}} f(z) dz \end{aligned}$$

由 Cauchy 定理, $\oint_C f(z) dz = 0$.

取 $R \rightarrow \infty$, $\delta_i \rightarrow 0$ ($i = 1, 2, 3$) 的极限, 则有:

$$\begin{aligned} \text{v.p.} \int_{-\infty}^{\infty} f(z) dz &= \int_{-\infty}^{-\delta_1} f(z) dz + \int_{\delta_1}^{1-\delta_2} f(z) dz + \int_{1+\delta_2}^{2-\delta_3} f(z) dz + \int_{2+\delta_3}^{\infty} f(z) dz \\ &= - \int_{C_R} f(z) dz - \int_{C_{\delta_1}} f(z) dz - \int_{C_{\delta_2}} f(z) dz - \int_{C_{\delta_3}} f(z) dz \end{aligned}$$

由大圆弧定理:

$$\lim_{R \rightarrow \infty} \int_{C_R} f(z) dz = 0$$

由小圆弧定理:

$$\begin{aligned} \lim_{\delta_1 \rightarrow 0} \int_{C_{\delta_1}} f(z) dz &= i\pi \lim_{z \rightarrow 0} z f(z) = \frac{i\pi}{2} \\ \lim_{\delta_2 \rightarrow 0} \int_{C_{\delta_2}} f(z) dz &= i\pi \lim_{z \rightarrow 1} (z-1) f(z) = -i\pi \\ \lim_{\delta_3 \rightarrow 0} \int_{C_{\delta_3}} f(z) dz &= i\pi \lim_{z \rightarrow 2} (z-2) f(z) = \frac{i\pi}{2} \end{aligned}$$

故:

$$\text{v.p.} \int_{-\infty}^{\infty} \frac{1}{z(z-1)(z-2)} dz = \frac{i\pi}{2} - i\pi + \frac{i\pi}{2} = 0$$

8.(4) 记 $f(z) = \frac{e^{pz}}{1-e^z}$, $I = \int_{-\infty}^{\infty} f(z) dz$.

考虑积分: $\oint_C f(z) dz$, 积分围道 C 如下:

从 $z = -R$ 沿实轴向右, 经过半圆 C_{δ_1} 绕过奇点 $z_1 = 0$ 到 $z = R$, 再沿 $\text{Re } z = R$ 到 $z = R + 2\pi i$, 再沿 $\text{Im } z = 2\pi$ 经过半圆 C_{δ_2} 绕过奇点 $z_2 = 2\pi i$ 到 $z = -R + 2\pi i$, 最后沿 $\text{Re } z = -R$ 回到 $z = -R$. 即:

$$\begin{aligned} \oint_C f(z) dz &= \int_{-R}^{-\delta_1} f(z) dz + \int_{C_{\delta_1}} f(z) dz + \int_{\delta_1}^R f(z) dz + \int_R^{R+2\pi i} f(z) dz \\ &\quad + \int_{R+2\pi i}^{\delta_2+2\pi i} f(z) dz + \int_{C_{\delta_2}} f(z) dz + \int_{-\delta_2+2\pi i}^{-R+2\pi i} f(z) dz + \int_{-R+2\pi i}^{-R} f(z) dz \end{aligned}$$

由 Cauchy 定理, $\oint_C f(z) dz = 0$.

取 $R \rightarrow \infty$, $\delta_i \rightarrow 0$ ($i = 1, 2$) 的极限, 则有:

$$\begin{aligned} I &= \int_{-R}^{-\delta_1} f(z) dz + \int_{\delta_1}^R f(z) dz \\ -e^{2\pi i} I &= \int_{R+2\pi i}^{\delta_2+2\pi i} f(z) dz + \int_{-\delta_2+2\pi i}^{-R+2\pi i} f(z) dz \end{aligned}$$

则有:

$$(1 - e^{2\pi i}) I = \int_{C_{\delta_1}} f(z) dz + \int_R^{R+2\pi i} f(z) dz + \int_{C_{\delta_2}} f(z) dz + \int_{-R+2\pi i}^{-R} f(z) dz$$

由于:

$$\left| \int_R^{R+2\pi i} f(z) dz \right| = \left| \int_0^{2\pi} \frac{e^{p(R+iy)}}{1 - e^{R+iy}} i dy \right| \leq \frac{e^{pR}}{e^R - 1} \int_0^{2\pi} dy = \frac{2\pi e^{pR}}{e^R - 1}$$

则:

$$\lim_{R \rightarrow \infty} \int_R^{R+2\pi i} f(z) dz = \lim_{R \rightarrow \infty} \frac{2\pi e^{pR}}{e^R - 1} = 0$$

由于:

$$\left| \int_{-R+2\pi i}^{-R} f(z) dz \right| = \left| \int_{2\pi}^0 \frac{e^{p(-R+iy)}}{1 - e^{-R+iy}} i dy \right| \leq e^{-pR} \int_{2\pi}^0 dy = 2\pi e^{-pR}$$

则:

$$\lim_{R \rightarrow \infty} \int_{-R+2\pi i}^{-R} f(z) dz = \lim_{R \rightarrow \infty} 2\pi e^{-pR} = 0$$

由小圆弧定理:

$$\begin{aligned} \lim_{\delta_1 \rightarrow 0} \int_{C_{\delta_1}} f(z) dz &= i\pi \lim_{z \rightarrow 0} z f(z) \\ &= i\pi \lim_{z \rightarrow 0} \frac{ze^{pz}}{1 - e^z} \\ &= i\pi \lim_{z \rightarrow 0} \frac{e^{pz} + pz e^{pz}}{-e^z} \\ &= -i\pi \end{aligned}$$

$$\begin{aligned} \lim_{\delta_2 \rightarrow 0} \int_{C_{\delta_2}} f(z) dz &= i\pi \lim_{z \rightarrow 2\pi i} (z - 2\pi i) f(z) \\ &= i\pi \lim_{z \rightarrow 2\pi i} \frac{(z - 2\pi i)e^{pz}}{1 - e^z} \\ &= i\pi \lim_{z \rightarrow 0} \frac{e^{pz} + p(z - 2\pi i)e^{pz}}{-e^z} \\ &= -i\pi e^{2p\pi i} \end{aligned}$$

即得:

$$\begin{aligned} (1 - e^{2p\pi i})I &= -i\pi(1 + e^{2p\pi i}) \\ I &= i\pi \frac{e^{2p\pi i} + 1}{e^{2p\pi i} - 1} = i\pi \frac{e^{p\pi i} + e^{-p\pi i}}{e^{p\pi i} - e^{-p\pi i}} = i\pi \frac{e^{pi} + e^{-pi}}{e^{pi} - e^{-pi}} = \pi \cot p \end{aligned}$$

故所求积分为:

$$\int_{-\infty}^{\infty} \frac{e^{pz} - e^{qz}}{1 - e^z} dz = \pi(\cot p - \cot q)$$

第八章

1.(1) 由 $\Gamma(z+1) = z\Gamma(z)$, $\Gamma(1) = 1$ 得: $\Gamma(n+1) = n!$ ($n \in \mathbb{Z}$).

则:

$$(2n)!! = 2^n n! = 2^n \Gamma(n+1)$$

1.(2) 由 $\Gamma(z+1) = z\Gamma(z)$, $\Gamma(1) = 1$ 得: $\Gamma(n+1) = n!$ ($n \in \mathbb{Z}$).

则:

$$(2n-1)!! = \frac{(2n)!}{2^n n!} = \frac{\Gamma(2n+1)}{2^n \Gamma(n+1)}$$

1.(3) 由 $\Gamma(z+1) = z\Gamma(z)$ 得:

$$(1+\nu)(2+\nu)\cdots(n+\nu) = \frac{\Gamma(n+1+\nu)}{\Gamma(1+\nu)}$$

1.(4) 由 $\Gamma(z+1) = z\Gamma(z)$ 及互余宗量定理 $\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin \pi z}$ 得:

$$\begin{aligned} & [n(n+1) - \nu(\nu+1)][(n-1)n - \nu(\nu+1)] \cdots [0 \cdot 1 - \nu(\nu+1)] \\ &= (n-\nu)(n+\nu+1)(n-1-\nu)(n+\nu) \cdots (-\nu)(\nu+1) \\ &= \frac{\Gamma(n-\nu+1)}{\Gamma(-\nu)} \frac{\Gamma(n+\nu+2)}{\Gamma(\nu+1)} \\ &= \frac{\Gamma(n-\nu+1)\Gamma(n+\nu+2)\Gamma(\nu-n)}{\Gamma(-\nu)\Gamma(\nu+1)\Gamma(\nu-n)} \\ &= \frac{\sin(-\pi\nu)}{\sin \pi(\nu-n)} \frac{\Gamma(n+\nu+2)}{\Gamma(\nu-n)} \\ &= -\frac{\sin \pi\nu}{\sin(\pi\nu - n\pi)} \frac{\Gamma(n+\nu+2)}{\Gamma(\nu-n)} \\ &= (-1)^{n+1} \frac{\Gamma(n+\nu+2)}{\Gamma(\nu-n)} \end{aligned}$$

2 令 $t = \ln \frac{1}{x} = -\ln x$, 则 $x = e^{-t}$, $dx = -e^{-t}dt$.

故:

$$\int_0^1 \left(\ln \frac{1}{x}\right)^{z-1} dx = \int_{\infty}^0 t^{z-1} (-e^{-t}) dt = \int_0^{\infty} e^{-t} t^{z-1} dt = \Gamma(z), \quad \operatorname{Re} z > 0$$

2.(1) 记 $f(z) = z^{-\alpha}e^{-z}$, 并规定正实轴上 $\arg z = 0$.

考虑积分: $\oint_C f(z)dz$, 积分围道 C 如下:

从 $z = \delta$ 沿实轴向右到 $z = R$, 经过四分之一圆弧 C_R 到 $z = iR$, 再沿虚轴向下到 $z = i\delta$, 再经过四分之一圆弧 C_δ 回到 $z = \delta$. 即:

$$\begin{aligned} \oint_C f(z)dz &= \int_{\delta}^R f(z)dz + \int_{C_R} f(z)dz + \int_{iR}^{i\delta} f(z)dz + \int_{C_\delta} f(z)dz \\ &= \int_{\delta}^R x^{-\alpha}e^{-x}dx + \int_{C_R} f(z)dz + \int_R^{\delta} \left(e^{\frac{i\pi}{2}}y\right)^{-\alpha} e^{iy}dy + \int_{C_\delta} f(z)dz \end{aligned}$$

由 Cauchy 定理, $\oint_C f(z)dz = 0$.

当 $\alpha < 1$ 时, 有:

$$\lim_{z \rightarrow 0} zf(z) = \lim_{z \rightarrow 0} z \cdot z^{-\alpha} e^{-z} = 0$$

由小圆弧定理:

$$\lim_{\delta \rightarrow 0} \int_{C_\delta} f(z)dz = 0$$

当 $\alpha > 0$ 时, 有:

$$\lim_{z \rightarrow \infty} z^{-\alpha} = 0$$

由 Jordan 引理:

$$\lim_{R \rightarrow \infty} \int_{C_R} f(z)dz = 0$$

取 $R \rightarrow \infty, \delta \rightarrow 0$ 的极限, 则有:

$$\int_0^0 \left(e^{\frac{i\pi}{2}} y\right)^{-\alpha} e^{iy} dy = e^{\frac{i(1-\alpha)\pi}{2}} \int_0^\infty y^{-\alpha} e^{-iy} dy = \int_0^\infty x^{-\alpha} e^{-x} dx = \Gamma(1-\alpha)$$

即:

$$\int_0^\infty x^{-\alpha} (\cos x - i \sin x) dx = \left[\cos \frac{(1-\alpha)\pi}{2} - i \sin \frac{(1-\alpha)\pi}{2} \right] \Gamma(1-\alpha)$$

分别比较实部和虚部, 得:

$$\begin{aligned} \int_0^\infty x^{-\alpha} \sin x dx &= \Gamma(1-\alpha) \cos \frac{\alpha\pi}{2} \\ \int_0^\infty x^{-\alpha} \cos x dx &= \Gamma(1-\alpha) \sin \frac{\alpha\pi}{2} \end{aligned}$$

又 $x^{-\alpha} \sin x$ 是 $0 < \alpha < 2$ 中的解析函数, 由解析函数的唯一性定理, 根据解析延拓原理可知:

$$\int_0^\infty x^{-\alpha} \sin x dx = \Gamma(1-\alpha) \cos \frac{\alpha\pi}{2}$$

在 $0 < \alpha < 2$ 中仍成立.

2.(2) 记 $f(z) = z^{\alpha-1} e^{-z}$, 并规定正实轴上 $\arg z = 0$.

考虑积分: $\oint_C f(z)dz$, 积分围道 C 如下:

从 $z = \delta$ 沿实轴向右到 $z = R$, 经过圆弧 C_R 到 $z = Re^{i\theta}$, 再沿直线 $\arg z = \theta$ 斜向下到 $z = \delta e^{i\theta}$, 再经过圆弧 C_δ 回到 $z = \delta$. 即:

$$\begin{aligned} \oint_C f(z)dz &= \int_\delta^R f(z)dz + \int_{C_R} f(z)dz + \int_{Re^{i\theta}}^{\delta e^{i\theta}} f(z)dz + \int_{C_\delta} f(z)dz \\ &= \int_\delta^R x^{\alpha-1} e^{-x} dx + \int_{C_R} f(z)dz + \int_R^\delta (re^{i\theta})^{\alpha-1} e^{-r(\cos\theta + i\sin\theta)} e^{i\theta} dr + \int_{C_\delta} f(z)dz \end{aligned}$$

由 Cauchy 定理, $\oint_C f(z)dz = 0$.

当 $\alpha > 0$ 时, 有:

$$\lim_{z \rightarrow 0} zf(z) = \lim_{z \rightarrow 0} z \cdot z^{\alpha-1} e^{-z} = 0$$

由小圆弧定理:

$$\lim_{\delta \rightarrow 0} \int_{C_\delta} f(z)dz = 0$$

当 $\alpha < 1$ 时, 有:

$$\lim_{z \rightarrow \infty} z^{\alpha-1} = 0$$

由 Jordan 引理:

$$\lim_{R \rightarrow \infty} \int_{C_R} f(z) dz = 0$$

取 $R \rightarrow \infty, \delta \rightarrow 0$ 的极限, 则有:

$$\int_0^\infty r^{\alpha-1} e^{-r(\cos \theta + i \sin \theta)} dr = e^{-i\alpha\theta} \Gamma(\alpha)$$

分别比较实部和虚部, 并将积分变量 r 换成 x , 得:

$$\begin{aligned} \int_0^\infty x^{\alpha-1} e^{-x \cos \theta} \sin(x \sin \theta) dx &= \sin(\alpha\theta) \Gamma(\alpha) \\ \int_0^\infty x^{\alpha-1} e^{-x \cos \theta} \cos(x \sin \theta) dx &= \cos(\alpha\theta) \Gamma(\alpha) \end{aligned}$$

3.(1) 由 $\Gamma(z+1) = z\Gamma(z)$ 得:

$$\ln \Gamma(z+1) = \ln z + \ln \Gamma(z)$$

两边求导, 由 $\psi(z) = \frac{d}{dz} \ln \Gamma(z)$ 得:

$$\psi(z+1) = \frac{1}{z} + \psi(z)$$

3.(2) 由 $\psi(z+1) = \frac{1}{z} + \psi(z)$, 将 z 换元为 $z+k-1$ 得:

$$\psi(z+k) - \psi(z+k-1) = \frac{1}{z+k-1}$$

两边对 k 求和得:

$$\begin{aligned} \sum_{k=0}^n \psi(z+k) - \sum_{k=0}^n \psi(z+k-1) &= \sum_{k=0}^n \frac{1}{z+k-1} \\ \psi(z+n) - \psi(z) &= \frac{1}{z} + \frac{1}{z+1} + \cdots + \frac{1}{z+n-1} \end{aligned}$$

3.(3) 由互余宗量定理 $\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin \pi z}$ 得:

$$\ln \Gamma(z) + \ln \Gamma(1-z) = \ln \pi - \ln(\sin \pi z)$$

两边求导, 由 $\psi(z) = \frac{d}{dz} \ln \Gamma(z)$ 得:

$$\begin{aligned} \psi(z) - \psi(1-z) &= \left(-\frac{1}{\sin \pi z} \right) \pi \cos \pi z = -\pi \cot \pi z \\ \psi(1-z) - \psi(z) &= \pi \cot \pi z \end{aligned}$$

3.(4) 由倍乘公式 $\Gamma(2z) = \frac{2^{2z-1}}{\sqrt{\pi}} \Gamma(z) \Gamma(z + \frac{1}{2})$ 得:

$$\ln \Gamma(2z) = (2z-1) \ln 2 - \frac{1}{2} \ln \pi + \ln \Gamma(z) + \ln \Gamma(z + \frac{1}{2})$$

两边求导, 由 $\psi(z) = \frac{d}{dz} \ln \Gamma(z)$ 得:

$$\begin{aligned} 2\psi(2z) &= 2 \ln 2 + \psi(z) + \psi(z + \frac{1}{2}) \\ 2\psi(2z) - \psi(z) - \psi(z + \frac{1}{2}) &= 2 \ln 2 \end{aligned}$$

第九章

1.(1) 由像函数导数的反演公式 $\mathcal{L}\{(-1)^n t^n f(t)\} = F^{(n)}(p)$ 及 $\mathcal{L}\{1\} = \frac{1}{p}$ 得:

$$\mathcal{L}\{(-1)^n t^n\} = \frac{d^n}{dp^n} \left(\frac{1}{p} \right) = (-1)^n \frac{n!}{p^{n+1}}$$

由于 Laplace 变换是线性变换, 则:

$$\mathcal{L}\{t^n\} = \frac{n!}{p^{n+1}}$$

1.(3) 由 Laplace 变换的位移定理 $\mathcal{L}\{e^{p_0 t} f(t)\} = F(p - p_0)$ 及 $\mathcal{L}\{\sin \omega t\} = \frac{\omega}{p^2 + \omega^2}$ 得:

$$\mathcal{L}\{e^{\lambda t} \sin \omega t\} = \frac{\omega}{(p - \lambda)^2 + \omega^2}$$

1.(5) 由 $\mathcal{L}\{1\} = \frac{1}{p}$ 及 $\mathcal{L}\{\cos \omega t\} = \frac{p}{p^2 + \omega^2}$ 得:

$$\mathcal{L}\{1 - \cos \omega t\} = \frac{1}{p} - \frac{p}{p^2 + \omega^2} = \frac{1}{p} - \frac{1}{2(p + i\omega)} - \frac{1}{2(p - i\omega)}$$

当 $t \rightarrow 0$ 时, 由 l'Hospital 法则得 $\left| \frac{1 - \cos \omega t}{t} \right| \rightarrow 0$.

由像函数积分的反演公式 $\int_p^\infty \mathcal{L}\{f(t)\} dt = \frac{f(t)}{t}$ 得:

$$\begin{aligned} \mathcal{L}\left\{\frac{1 - \cos \omega t}{t}\right\} &= \int_p^\infty \left(\frac{1}{p} - \frac{1}{2(p + i\omega)} - \frac{1}{2(p - i\omega)} \right) dt \\ &= \ln p - \frac{1}{2} \ln(p + i\omega) - \frac{1}{2} \ln(p - i\omega) \Big|_p^\infty \\ &= \frac{1}{2} \ln \frac{p^2}{p^2 + \omega^2} \Big|_p^\infty \\ &= \frac{1}{2} \ln \frac{p^2 + \omega^2}{p^2} \end{aligned}$$

最后一步使用了 l'Hospital 法则.

当 $t \rightarrow 0$ 时, 使用两次 l'Hospital 法则得 $\left| \frac{1 - \cos \omega t}{t^2} \right| \rightarrow \frac{\omega^2}{2}$.

由像函数积分的反演公式 $\int_p^\infty \mathcal{L}\{f(t)\} dt = \frac{f(t)}{t}$ 得:

$$\begin{aligned} \mathcal{L}\left\{\frac{1 - \cos \omega t}{t^2}\right\} &= \int_p^\infty \frac{1}{2} \ln \frac{p^2 + \omega^2}{p^2} dt \\ &= -\frac{p}{2} \ln \frac{p^2 + \omega^2}{p^2} + \omega \arctan \frac{\omega}{p} \end{aligned}$$

3.(1) $f(t) = |\sin \omega t|$ 为周期函数, 周期为 $a = \frac{\pi}{\omega}$. 则:

$$\begin{aligned}\mathcal{L}\{f(t)\} &= \int_0^{\infty} e^{-pt} f(t) dt \\ &= \sum_{n=0}^{\infty} \int_{na}^{(n+1)a} e^{-pt} f(t) dt \\ &= \sum_{n=0}^{\infty} \int_0^a e^{-p(t+na)} f(t) dt \\ &= \sum_{n=0}^{\infty} (e^{-pa})^n \int_0^a e^{-pt} f(t) dt \\ &= \frac{1}{1 - e^{-pa}} \int_0^a e^{-pt} f(t) dt\end{aligned}$$

最后一步使用了 $\sum_{n=0}^{\infty} z^n = \frac{1}{1-z}$.

又因为:

$$\begin{aligned}\int_0^a e^{-pt} |\sin \omega t| dt &= \int_0^a e^{-pt} \sin \omega t dt \\ &= \int_0^a e^{-pt} \left(\frac{e^{i\omega t} - e^{-i\omega t}}{2i} \right) dt \\ &= (1 + e^{-\frac{\pi p}{\omega}}) \frac{\omega}{p^2 + \omega^2}\end{aligned}$$

则:

$$\mathcal{L}\{|\sin \omega t|\} = \frac{(1 + e^{-\frac{\pi p}{\omega}})}{(1 - e^{-\frac{\pi p}{\omega}})} \frac{\omega}{p^2 + \omega^2} = \frac{\omega}{p^2 + \omega^2} \coth \frac{\pi p}{2\omega}$$

4.(1) 记 $F(p) = \frac{a^3}{p(p+a)^3}$.

对于 $G(p) = e^{pt} F(p) = \frac{e^{pt} a^3}{p(p+a)^3}$, 函数的奇点为 $p_1 = 0, p_2 = -a$.

对于 $p_1 = 0$, $G(p) = \frac{e^{pt} a^3}{p(p+a)^3} = \frac{P(z)}{Q(z)}$, $P(0) = a^3 \neq 0$, $Q(0) = 0$, $Q'(0) = a^3 \neq 0$, 则 $\text{res}G(0) = \frac{P(0)}{Q'(0)} = 1$.

对于 $p_2 = -a$, p_2 为 $G(p)$ 的三阶极点

$$\begin{aligned}\text{res}f(-a) &= \frac{1}{(3-1)!} \lim_{p \rightarrow -a} \left[\frac{d^2}{dp^2} (p+a)^3 \frac{e^{pt} a^3}{p(p+a)^3} \right] \\ &= \frac{1}{2} \lim_{p \rightarrow -a} \left[(p^2 t^2 - 2pt + 2) e^{pt} \frac{a^3}{p^3} \right] \\ &= - \left(\frac{1}{2} a^2 t^2 + at + 1 \right) e^{-at}\end{aligned}$$

由 $f(t) = \sum \text{res}\{e^{pt} F(p)\} \eta(t)$ 得:

$$f(t) = \sum \text{res}\{G(p)\} \eta(t) = \left[1 - \left(\frac{1}{2} a^2 t^2 + at + 1 \right) e^{-at} \right] \eta(t)$$

4.(3) 记 $F(p) = \frac{4p-1}{(p^2+p)(4p^2-1)}$.

对于 $G(p) = e^{pt} F(p) = \frac{e^{pt}(4p-1)}{(p^2+p)(4p^2-1)}$, 函数的奇点为 $p_1 = 0, p_2 = 1, p_3 = \frac{1}{2}, p_4 = -\frac{1}{2}$, 均为一阶极点.

对于 $p_1 = 0, G(p) = \frac{e^{pt}(4p-1)}{(p^2+p)(4p^2-1)} = \frac{P(z)}{Q(z)}, P(0) = -1 \neq 0, Q(0) = 0, Q'(0) = -1 \neq 0$, 则 $\text{res}G(0) = \frac{P(0)}{Q'(0)} = 1$.

同理可得:

$$\text{res}G(1) = \frac{P(1)}{Q'(1)} = \frac{5}{3}e^{-t}$$

$$\text{res}G\left(\frac{1}{2}\right) = \frac{P\left(\frac{1}{2}\right)}{Q'\left(\frac{1}{2}\right)} = \frac{1}{3}e^{\frac{1}{2}t}$$

$$\text{res}G\left(-\frac{1}{2}\right) = \frac{P\left(-\frac{1}{2}\right)}{Q'\left(-\frac{1}{2}\right)} = -3e^{\frac{1}{2}t}$$

由 $f(t) = \sum \text{res}\{e^{pt}F(p)\}\eta(t)$ 得:

$$f(t) = \sum \text{res}\{G(p)\}\eta(t) = \left(1 + \frac{5}{3}e^{-t} + \frac{1}{3}e^{\frac{1}{2}t} - 3e^{\frac{1}{2}t}\right)\eta(t)$$

4.(5) 由像函数导数的反演公式 $\mathcal{L}\{(-1)^n t^n f(t)\} = F^{(n)}(p)$ 及 $\mathcal{L}\{1\} = \frac{1}{p}$ 得:

$$\mathcal{L}\{(-1)^n t^n\} = \frac{d^n}{dp^n}\left(\frac{1}{p}\right) = (-1)^n \frac{n!}{p^{n+1}}$$

由于 Laplace 变换是线性变换, 则:

$$\mathcal{L}\{t^n\} = \frac{n!}{p^{n+1}}$$

则有:

$$\mathcal{L}\{t\} = \frac{1}{p^2}$$

由 Laplace 变换的延迟定理 $\mathcal{L}\{f(t-\tau)\} = e^{-p\tau}F(p)$ 得:

$$\mathcal{L}\{t-\tau\} = \frac{e^{-p\tau}}{p^2}$$

故 $\frac{e^{-p\tau}}{p^2}$ 的原函数为 $(t-\tau)\eta(t-\tau)$

5.(2) 设流过电感 L 的电流为 $i_1(t) = i(t)$, 流过电容 C 的电流为 $i_2(t)$.

由节点电流方程, 流过电阻 R 的电流为 $i_1(t) - i_2(t)$.

由回路电压方程, 对左侧回路:

$$E - L \frac{d}{dt} i_1(t) - \frac{1}{C} \int_0^t i_2(t) dt = 0$$

对右侧回路:

$$\frac{1}{C} \int_0^t i_2(t) dt - [i_1(t) - i_2(t)] R = 0$$

对两式等号两边进行 Laplace 变换得:

$$\begin{cases} \frac{E}{p} - LpI_1(p) - \frac{1}{C} \frac{I_2(p)}{p} = 0 \\ \frac{1}{C} \frac{I_2(p)}{p} - [I_1(p) - I_2(p)] R = 0 \end{cases} \quad (1)$$

其中 $I_1(p) = \mathcal{L}\{i_1(t)\}$, $I_2(p) = \mathcal{L}\{i_2(t)\}$.

联立解得:

$$\begin{cases} I_1 = \frac{EC(1+pCR)}{LCp^2(1+pCR)+pCR} \\ I_2 = \frac{EC(pCR)}{LCp^2(1+pCR)+pCR} \end{cases} \quad (2)$$

代入初始条件 $i(0) = 0$, $q(0) = 0$, 对 I_1 进行反演得:

$$i(t) = i_1(t) = \begin{cases} \frac{E}{R} (1 + Ae^{-\gamma_1 t} - Be^{-\gamma_2 t}) & k < 1 \\ \frac{E}{R} (1 - e^{-\alpha t} - \frac{\alpha t}{2} e^{-\alpha t}) & k = 1 \\ \frac{E}{R} [1 - e^{-\alpha t} (\cos \omega t + \frac{\alpha - \beta}{\omega} \sin \omega t)] & k > 1 \end{cases} \quad (3)$$

其中 $\alpha = \frac{1}{2RC}$, $\beta = \frac{R}{L}$, $k = 4R^2 \frac{C}{L}$, $\gamma_1 = \alpha(1 - \sqrt{1 - k})$, $\gamma_2 = \alpha(1 + \sqrt{1 - k})$, $A = \frac{\beta - \gamma_2}{\gamma_2 - \gamma_1}$, $B = \frac{\beta - \gamma_1}{\gamma_2 - \gamma_1}$.

6.(1) 由像函数积分的反演公式 $\int_0^\infty \frac{f(t)}{t} dt = \int_0^\infty F(p) dp$ 得:

$$\begin{aligned} \int_0^\infty \frac{e^{-ax} - e^{-bx}}{x} \cos cx dx &= \int_0^\infty \mathcal{L} \{ (e^{-ax} - e^{-bx}) \cos cx \} dp \\ &= \int_0^\infty \mathcal{L} \left\{ (e^{-ax} - e^{-bx}) \frac{e^{icx} - e^{-icx}}{2} \right\} dp \\ &= \int_0^\infty \frac{1}{2} \mathcal{L} \{ e^{(-a+ic)x} - e^{(-b+ic)x} + e^{(-a-ic)x} - e^{(-b-ic)x} \} dp \\ &= \int_0^\infty \frac{1}{2} \left(\frac{1}{p+a-ic} - \frac{1}{p+b-ic} + \frac{1}{p+a+ic} - \frac{1}{p+b+ic} \right) dp \\ &= \frac{1}{2} \ln \frac{(p+a-ic)(p+a+ic)}{(p+b-ic)(p+b+ic)} \Big|_0^\infty \\ &= \frac{1}{2} \ln \frac{(p+a)^2 + c^2}{(p+b)^2 + c^2} \Big|_0^\infty \\ &= \frac{1}{2} \ln \frac{b^2 + c^2}{a^2 + c^2} \end{aligned}$$

6.(3) 不妨设 $t > 0$.

由含参数积分所定义函数的 Laplace 变换 $\mathcal{L} \{ \int_0^\infty f(t, \tau) d\tau \} = \int_0^\infty \mathcal{L} \{ f(t, \tau) \} d\tau$ 得:

$$\begin{aligned} \mathcal{L} \left\{ \int_0^\infty \frac{\sin xt}{x(x^2+1)} dx \right\} &= \int_0^\infty \mathcal{L} \left\{ \frac{\sin xt}{x(x^2+1)} \right\} dx \\ &= \int_0^\infty \frac{1}{x(x^2+1)} \frac{x}{p^2+x^2} dx \\ &= \int_0^\infty \frac{1}{(x^2+1)(x^2+p^2)} dx \\ &= \frac{1}{2} \int_{-\infty}^\infty \frac{1}{(x^2+1)(x^2+p^2)} dx \end{aligned}$$

考虑积分: $\oint_C \frac{1}{(z^2+1)(z^2+p^2)} dz$, 积分围道 C 如下:

从 $z = -R$ 沿实轴到 $z = R$, 再沿 $|z| = R$, $0 < \arg z < \pi$ 的半圆 C_R 逆时针回到 $z = -R$. 即:

$$\oint_C \frac{1}{(z^2+1)(z^2+p^2)} dz = \int_{-R}^R \frac{1}{(z^2+1)(z^2+p^2)} dz + \int_{C_R} \frac{1}{(z^2+1)(z^2+p^2)} dz$$

取 $R \rightarrow \infty$ 的极限, 则有:

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{1}{(x^2+1)(x^2+p^2)} dx &= \int_{-\infty}^{\infty} \frac{1}{(z^2+1)(z^2+p^2)} dz \\ &= \lim_{R \rightarrow \infty} \oint_C \frac{1}{(z^2+1)(z^2+p^2)} dz - \lim_{R \rightarrow \infty} \int_{C_R} \frac{1}{(z^2+1)(z^2+p^2)} dz \end{aligned}$$

由大圆弧定理:

$$\lim_{R \rightarrow \infty} \int_{C_R} \frac{1}{(z^2+1)(z^2+p^2)} dz = 0$$

故:

$$\int_{-\infty}^{\infty} \frac{1}{(x^2+1)(x^2+p^2)} dx = \lim_{R \rightarrow \infty} \oint_C \frac{1}{(z^2+1)(z^2+p^2)} dz = 2\pi i \sum_{\text{Im } z > 0} \text{res} \frac{1}{(z^2+1)(z^2+p^2)}$$

在 $\text{Im } z > 0$ 的区域内, $F(z) = \frac{1}{(z^2+1)(z^2+p^2)}$ 有两个奇点 $z_1 = i$, $z_2 = pi$.

对于 $z_1 = i$, $F(z_1) = \frac{1}{(z^2+1)(z^2+p^2)} = \frac{P(z)}{Q(z)}$, $P(z_1) = 1 \neq 0$, $Q(z_1) = 0$, $Q'(z_1) = 2i(p^2-1) \neq 0$, 则 $\text{res} f(z_1) = \frac{P(z_1)}{Q'(z_1)} = \frac{1}{2i(p^2-1)}$.

对于 $z_2 = pi$, $F(z_2) = \frac{1}{(z^2+1)(z^2+p^2)} = \frac{P(z)}{Q(z)}$, $P(z_2) = 1 \neq 0$, $Q(z_2) = 0$, $Q'(z_2) = -2pi(p^2-1) \neq 0$, 则 $\text{res} f(z_2) = \frac{P(z_2)}{Q'(z_2)} = -\frac{1}{2pi(p^2-1)}$.

故:

$$\begin{aligned} \int_0^{\infty} \frac{1}{(x^2+1)(x^2+p^2)} dx &= \frac{1}{2} \int_{-\infty}^{\infty} \frac{1}{(x^2+1)(x^2+p^2)} dx \\ &= \pi i \sum_{\text{Im } z > 0} \text{res} \frac{1}{(z^2+1)(z^2+p^2)} \\ &= \pi i \left(\frac{1}{2i(p^2-1)} - \frac{1}{2pi(p^2-1)} \right) \\ &= \pi i \times \frac{1}{2pi(p+1)} \\ &= \frac{\pi}{2p(p+1)} \\ &= \frac{\pi}{2} \left(\frac{1}{p} - \frac{1}{p+1} \right) \end{aligned}$$

求反演得:

$$\int_0^{\infty} \frac{\sin xt}{x(x^2+1)} dx = \frac{\pi}{2} (1 - e^{-t})$$

考虑 $t \in \mathbb{R}$ 的情况, 则有:

$$\int_0^{\infty} \frac{\sin xt}{x(x^2+1)} dx = \frac{\pi}{2} (1 - e^{-|t|}) \text{sgn}(t)$$

第十章

2.(1) 当 $x \neq t$ 时, $\delta(x-t) = 0$.

分段求得:

$$g(x; t) = \begin{cases} A(t) \sinh kx + B(t) \cosh kx, & x < t \\ C(t) \sinh kx + D(t) \cosh kx, & x > t \end{cases} \quad (4)$$

根据初值条件, 可以定出:

$$A(t) = 0$$

$$B(t) = 0$$

由 $g(x; t)$ 在 $x = t$ 点连续得:

$$g(t-0; t) = g(t+0; t)$$

则:

$$C(t) \sinh kt + D(t) \cosh kt = 0$$

将方程两边从 $t-0$ 到 $t+0$ 积分, 得:

$$\left. \frac{d}{dx} g(x; t) \right|_{t-0}^{t+0} = 1$$

于是有:

$$kC(t) \cosh kx + kD(t) \sinh kx = 1$$

解得:

$$C(t) = \frac{1}{k} \cosh kt$$

$$D(t) = \frac{1}{k} \sinh kt$$

最终得:

$$g(x; t) = \frac{1}{k} \sinh k(x-t) \eta(x-t)$$

3.(2) 设 $y(x) = u(x) + v(x)$.

由 $u(x)$ 满足:

$$\frac{d^2}{dx^2} u(x) - k^2 u(x) = f(x), \quad x > 0, \quad k > 0$$

$$u(0) = 0, \quad \frac{d}{dx} u(0) = 0$$

得:

$$u(x) = \frac{1}{k} \int_0^x f(t) \sinh k(x-t) dt$$

则 $v(x)$ 需满足:

$$\frac{d^2}{dx^2} v(x) - k^2 v(x) = 0, \quad x > 0, \quad k > 0$$

$$v(0) = A, \quad \frac{d}{dx} v(0) = B$$

解得:

$$v(x) = A \cosh kx + \frac{B}{k} \sinh kx$$

最终得:

$$y(x) = A \cosh kx + \frac{B}{k} \sinh kx + \frac{1}{k} \int_0^x f(t) \sin k(x-t) dt$$